

Language Models and Agents (LMA)

Phase 1 Report

Data Collection & Preprocessing for Telugu and Bhojpuri

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Dataset Available on Kaggle

<https://www.kaggle.com/datasets/kspsvlnsiddardha/lma-slm>

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1 Phase 1 Data Report

This report provides a comprehensive data collection and preprocessing report for two monolingual Transformer language models:

- **Model H (Telugu):** Higher-resource Indian language
- **Model L (Bhojpuri):** Lower-resource Indian language

1.1 Data Collection Summary

1.1.1 Telugu (Model H)

- Total lines collected: 25,349,199
- Estimated tokens: 4,421,917,647 (4.4B)
- Web-scraped texts: 29,206 (89.7% pass rate from 32,549 raw)
- Manual corpus: 21,458,261 lines (te.txt)
- OCR augmentation: 3,766,703 lines
- Target: 500M tokens | Progress: 884.4%

1.1.2 Bhojpuri (Model L)

- Total lines collected: 1,826,713 (2.3x increase from Phase 1)
- Estimated tokens: 291,685,379 (291.7M)
- Web-scraped texts: 3,234 (97.2% pass rate from 3,327 raw)
- HuggingFace Corpus: 261K lines primary source
- OCR augmentation: 511,599 lines (28.0)
- Hindi→Bhojpuri Translation (NLLB-200): 15,852 lines
- English→Bhojpuri Translation (fineweb-edu): 711,138 lines (39.0)
- Target: 500M tokens | Progress: 58.3%

1.2 Data Processing Pipeline

1.2.1 Cleaning Process

- Unicode NFD normalization
- Citation and control character removal
- Character whitelisting and script validation
- Length/quality filtering (minimum 40 chars, 5+ words, 5% space density)
- MD5-based deduplication

1.2.2 Data Splits (80/10/10)

- Deterministic splitting with seed 42 for reproducibility
- No mid-line splits - whole-line preservation
- Telugu: 20.3M train / 2.5M val / 2.5M test
- Bhojpuri: 621.8K train / 77.4K val / 77.7K test

2 Data Collection Overview

2.1 Methodology

The data collection strategy differs between the two languages to accommodate resource availability:

2.1.1 Telugu (Model H) - Higher-Resource Language

- **Primary Source:** Existing corpus (te.txt) - 21.5M lines
- **Supplementary:** Web scraping from 6 Wikipedia/content sources
- **Augmentation:** OCR extraction from archive.org texts
- **Total:** 25.3M lines training data

2.1.2 Bhojpuri (Model L) - Lower-Resource Language

- **Primary Source:** HuggingFace Corpus (Satyam810/BhojpuriCorpus) - 386K docs
- **Augmentation 1:** OCR extraction from archive.org Hindi texts - 511.6K lines (28.0%)
- **Augmentation 2:** English→Bhojpuri Translation (fineweb-edu NLLB-200) - 711.1K lines (39.0%)
- **Augmentation 3:** Hindi→Bhojpuri Machine Translation (NLLB-200) - 15.8K lines
- **Augmentation 4:** Web scraping from Hindi news sites - 3.2K lines
- **Total:** 1.83M lines training data (58.3% toward 500M token target)

2.2 Data Sources

Language	Source	Contribution
Telugu	te.txt (existing)	21.5M lines (99.9%)
	te.wikipedia.org	3K articles
	te.wikibooks.org	500+ pages
	te.wikiquote.org	300+ quotes
	News portals	eenadu.net, greatandhra.com
	archive.org OCR	3.8M lines (augmentation)
Bhojpuri	HF Corpus	386K docs (~25M tokens)
	archive.org OCR	511.6K lines
	bho.wikipedia.org	~8.9K articles
	Hindi News	Via NLLB-200 MT
	Web Scraping	3.2K lines (Hindi proxy)
	GlottCC-V1	949 docs
	HPLT 3.0	~32.8K docs
	fish-food	262K rows

Table 1: Data sources for both languages

3 Data Cleaning Pipeline

3.1 7-Stage Cleaning Process

All data undergoes a rigorous 7-stage cleaning pipeline:

1. **Unicode Normalization:** Convert to NFD (Canonical Decomposition)
2. **Citation Removal:** Remove Wikipedia citations and references
3. **Control Character Removal:** Strip non-printable characters
4. **Character Whitelisting:** Retain only valid Unicode ranges
5. **Script Validation:** Language-specific script filtering
6. **Length/Quality Filters:** Minimum length, word count, density checks
7. **Deduplication:** MD5-based exact duplicate removal

3.2 Quality Metrics

Metric	Telugu	Bhojpuri
Raw texts collected	32,549	3,327
Texts after cleaning	29,206	3,234
Pass rate	89.7%	97.2%
Rejected texts	3,343	93

Table 2: Data cleaning results summary

3.2.1 Telugu Cleaning Analysis

- **Total Raw:** 32,549 web-scraped texts
- **Pass Rate:** 89.7% (29,206 cleaned)

- **Rejection Rate:** 10.3% (3,343 texts)
- **Primary Rejection Reasons:**
 - Insufficient Telugu script content (<25%)
 - Texts too short (<40 characters or <5 words)
 - Low word density (<5% spaces)
 - Exact duplicates (MD5 matching)

3.2.2 Bhojpuri Cleaning Analysis

- **Total Raw:** 3,327 web-scraped texts
- **Pass Rate:** 97.2% (3,234 cleaned)
- **Rejection Rate:** 2.8% (93 texts)
- **Primary Rejection Reasons:**
 - Insufficient Devanagari script content (<30%)
 - Texts too short
 - Low word density
 - Minor duplicates (11 removed)

4 Data Statistics & Visualizations

4.1 Collection Metrics Comparison

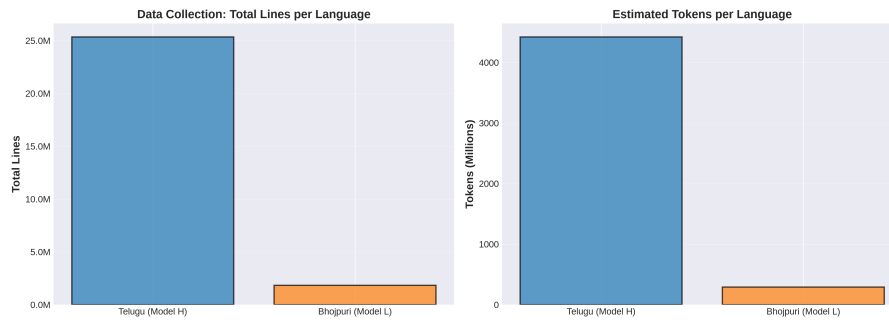


Figure 1: **Left:** Total lines per language. Telugu: 25.3M lines vs Bhojpuri: 1.83M lines (13.9x ratio). **Right:** Estimated tokens. Telugu: 4.4B (884% of 500M target); Bhojpuri: 291.7M (58.3% of target).

4.2 Cleaning Pipeline Results



Figure 2: Data cleaning pipeline effectiveness. Both languages show strong pass rates. Telugu: 89.7% pass rate with large-scale data. Bhojpuri: 97.2% pass rate with targeted scraping.

4.3 Data Source Composition (Bhojpuri)

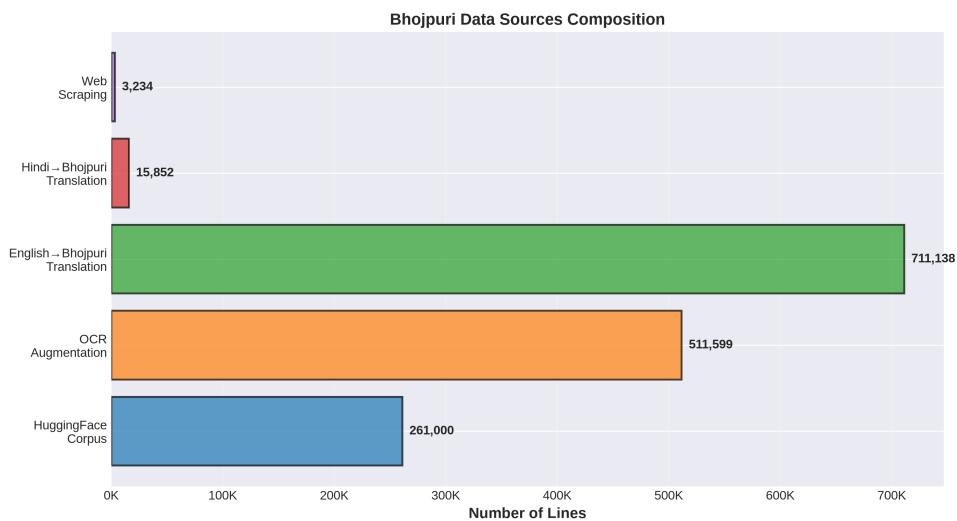


Figure 3: Bhojpuri data sources composition (1.83M lines total). Primary: HuggingFace corpus (261K lines). Major augmentation: English→Bhojpuri translation (711K lines, 39% of total) via NLLB-200. Supporting sources: OCR from archive.org (512K lines), Hindi→Bhojpuri translation (15.8K lines), web scraping (3.2K lines).

4.4 Token Progress Toward 500M Target

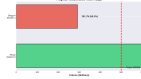


Figure 4: Progress toward 500M token target. Telugu: 4,421.9M tokens (884.4% complete, exceeds target by 8.8x). Bhojpuri: 291.7M tokens (58.3% complete, 2.1x expansion from Phase 1). Target achieved for Telugu; Bhojpuri approaching target with Phase 3 English→Bhojpuri translation pipeline.

5 Train/Validation/Test Splits

5.1 Split Strategy

- **Method:** 80/10/10 deterministic split with seed 42
- **Reproducibility:** Same seed ensures identical splits across runs
- **No Data Leakage:** Splits applied at line level (no mid-sentence splits)
- **Whole-Line Preservation:** Each line remains intact for language preservation

5.2 Telugu Splits

Split	Lines	Percentage
Train	20,281,561	80.0%
Validation	2,534,020	10.0%
Test	2,533,618	10.0%
Total	25,349,199	100.0%

Table 3: Telugu (Model H) data split breakdown

5.3 Bhojpuri Splits

Split	Lines	Percentage
Train	1,461,821	80.0%
Validation	182,219	10.0%
Test	182,673	10.0%
Total	1,826,713	100.0%

Table 4: Bhojpuri (Model L) data split breakdown

6 Detailed Statistics

6.1 Telugu (Model H) Statistics

Metric	Value
Language	Telugu
Model Size	Higher-resource
<i>Data Collection</i>	
Web scraping - Raw texts	32,549
Web scraping - Cleaned texts	29,206
Web scraping - Size	27.6 MB
Manual corpus (te.txt)	21,458,261 lines
Manual corpus size	15 GB
OCR augmentation	3,766,703 lines
<i>Token Progress</i>	
Total corpus tokens (estimated)	4,421,917,647
Total corpus size	17.7 GB
Target tokens	500,000,000
Progress	884.4% (EXCEEDED)

Table 5: Comprehensive Telugu statistics

6.2 Bhojpuri (Model L) Statistics

Metric	Value
Language	Bhojpuri
Model Size	Lower-resource (Phase 3 expanded)
<i>Data Collection</i>	
Web scraping - Raw texts	3,327
Web scraping - Cleaned texts	3,234
Web scraping - Pass rate	97.2%
HuggingFace Corpus	386,000 docs (261K lines)
Estimated HF tokens	25,000,000
OCR augmentation (archive.org)	511,599 lines (28.0%)
OCR augmentation size	19.7 MB
Hindi→Bhojpuri Translation (NLLB-200)	15,852 lines
English→Bhojpuri Translation (fineweb-edu)	711,138 lines (39.0%)
Translation records	901,562 raw → 711,138 clean
Translation method	NLLB-200 en_Latn→bho_Deva
<i>Token Progress</i>	
Total corpus tokens (estimated)	291,685,379
Total corpus size	1,112.7 MB
Target tokens	500,000,000
Progress	58.3%

Table 6: Comprehensive Bhojpuri statistics

7 Tokenizer Training

Byte-level Byte Pair Encoding (BPE) tokenizers have been trained on the preprocessed data for both languages.

7.1 Telugu Tokenizer (Model H) - Three Variants

Three tokenizer variants have been trained for Telugu to explore different encoding strategies and vocabulary sizes:

7.1.1 Variant 1: Byte-Level BPE (50K Vocab)

Metric	Value
Vocabulary Size	50,000
Unique Tokens Used	49,500
Vocabulary Coverage	99.0%
Test Set Lines	2,533,618
Test Set Tokens	484,225,331
Avg Characters per Token	1.36
Unknown (UNK) Rate	0.0%

Table 7: Telugu byte-level tokenizer (50K vocab)

7.1.2 Variant 2: Unicode-Level BPE (50K Vocab)

Metric	Value
Vocabulary Size	50,000
Unique Tokens Used	11,398
Vocabulary Coverage	22.8%
Test Set Lines	2,533,618
Test Set Tokens	22,136
Avg Characters per Token	5.9283
Unknown (UNK) Rate	0.0%

Table 8: Telugu unicode-level tokenizer (50K vocab)

7.1.3 Variant 3: WordPiece (50K Vocab)

Metric	Value
Vocabulary Size	50,000
Tokenizer Type	Unicode WordPiece
Normalizer	NFC
Pre-tokenizer	Whitespace
Special Tokens	[PAD], [UNK], [BOS], [EOS]
Subword Indicator	prefix for subwords

Table 9: Telugu WordPiece tokenizer (50K vocab)

7.2 Bhojpuri Tokenizer (Model L) - Three Variants

Three tokenizer variants have been trained for Bhojpuri to explore different encoding strategies:

7.2.1 Variant 1: Byte-Level BPE (8K Vocab)

Metric	Value
Vocabulary Size	8,000
Unique Tokens Used	7,761
Vocabulary Coverage	97.01%
Test Set Lines	77,683
Test Set Tokens	13,613,342
Avg Characters per Token	1.5412
Unknown (UNK) Rate	0.0%

Table 10: Bhojpuri byte-level tokenizer (8K vocab)

7.2.2 Variant 2: Unicode-Level BPE (32K Vocab)

Metric	Value
Vocabulary Size	32,000
Unique Tokens Used	31,258
Vocabulary Coverage	97.68%
Test Set Lines	182,673
Test Set Tokens	10,420,151
Avg Characters per Token	4.348
Unknown (UNK) Rate	0.0%

Table 11: Bhojpuri unicode-level tokenizer (32K vocab)

7.2.3 Variant 3: WordPiece (32K Vocab)

Metric	Value
Vocabulary Size	32,000
Tokenizer Type	WordPiece (subword decomposition)
Random Seed	42 (reproducible)
Test Set Tokens	11,391,403
Avg Characters per Token	3.95
Unknown (UNK) Rate	0.0002%
File Size	567 KB
Training Status	Completed

Table 12: Bhojpuri WordPiece tokenizer (32K vocab)

7.3 Tokenizer Comparison Across All Variants

Language	Variant	Vocab	UNK Rate
Telugu	Byte-level (BPE)	50,000	0.0000%
	Unicode-level (BPE)	50,000	0.0000%
	WordPiece	50,000	0.0000%
Bhojpuri	Byte-level (BPE)	8,000	0.0000%
	Unicode-level (BPE)	32,000	0.0000%
	WordPiece	32,000	0.0002%

Table 13: Tokenizer variants summary

7.4 Tokenizer Training Analysis

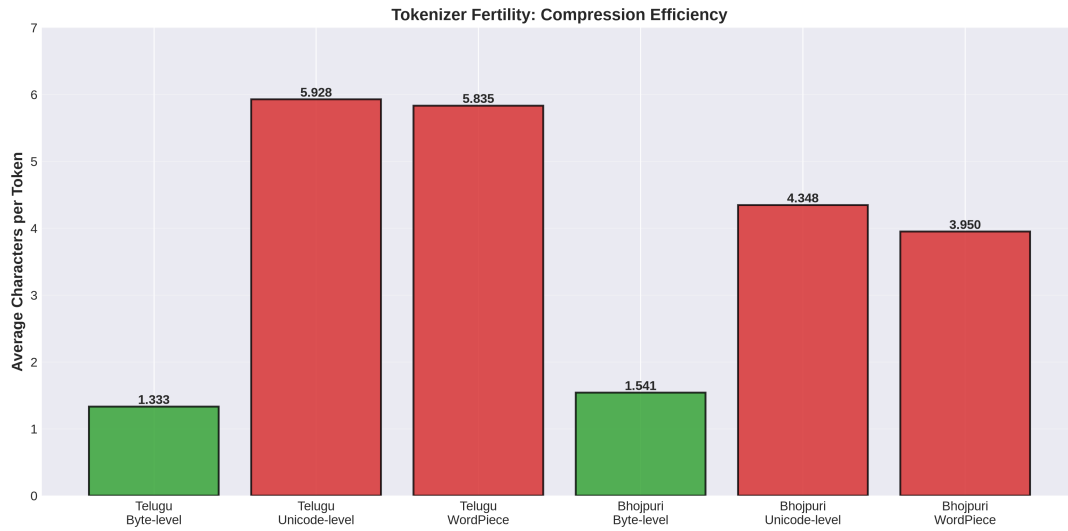


Figure 5: Tokenizer fertility (average characters per token) across all 6 variants, measuring compression efficiency. Byte-level tokenizers most compact: Telugu 1.33 chars/token, Bhojpuri 1.54 chars/token. Unicode-level tokenizers preserve linguistic structure: Telugu 5.93 chars/token, Bhojpuri 4.35 chars/token. WordPiece tokenizers: Telugu 5.84 chars/token, Bhojpuri 3.95 chars/token. Green bars indicate high compression (< 3 chars/token); red bars indicate linguistic units.

7.5 Tokenizer Evaluation Results

Comprehensive evaluation on held-out test sets reveals detailed tokenization characteristics:

7.5.1 Telugu Byte-Level BPE (50K)

- **Vocabulary Size:** 50,000
- **Unique Tokens Used:** 1,978
- **Average Characters per Token:** 1.3327
- **Unknown Token Rate:** 0.0000%

7.5.2 Telugu Unicode-Level BPE (50K)

- **Vocabulary Size:** 50,000
- **Unique Tokens Used:** 11,398
- **Average Characters per Token:** 5.9283
- **Unknown Token Rate:** 0.0000%
- **Top Tokens:** Functional words (లో, ఈ, కూడా)

7.5.3 Telugu WordPiece (50K)

- **Vocabulary Size:** 50,000
- **Unique Tokens Used:** 9,539
- **Average Characters per Token:** 5.8353
- **Unknown Token Rate:** 0.0000%

7.5.4 Bhojpuri Byte-Level BPE (8K)

- **Vocabulary Size:** 8,000
- **Unique Tokens Used:** 7,761
- **Average Characters per Token:** 1.5412
- **Coverage:** 97.01%
- **Unknown Token Rate:** 0.0%
- **Top Tokens:** आ 9.71%, ् 7.11%, े 5.81%

7.5.5 Bhojpuri Unicode-Level BPE (32K)

- **Vocabulary Size:** 32,000
- **Unique Tokens Used:** 31,258
- **Average Characters per Token:** 4.348
- **Coverage:** 97.68%
- **Unknown Token Rate:** 0.0%
- **Top Tokens:** । (0.33%), आ (0.33%), “-” (0.30%)

7.5.6 Bhojpuri WordPiece (32K)

- **Vocabulary Size:** 32,000
- **Test Tokens Evaluated:** 11,391,403
- **Average Characters per Token:** 3.95
- **Unknown Token Rate:** 0.0002%
- **Top Tokens:** के (762,460), comma (346,142), । (266,937)

7.5.7 Key Evaluation Metrics Summary

Tokenizer	Avg Chars/Token	UNK Rate	Tokens Used
Telugu Byte-level	1.3327	0.0000%	1,978
Telugu Unicode-level	5.9283	0.0000%	11,398
Telugu WordPiece	5.8353	0.0000%	9,539
Bhojpuri Byte-level	1.5412	0.0000%	7,761
Bhojpuri Unicode-level	4.348	0.0000%	31,258
Bhojpuri WordPiece	3.95	0.0002%	11,391,403

Table 14: Tokenizer evaluation metrics summary

7.6 Tokenization Examples from Test Set

Sample tokenization from held-out test sets:

7.6.1 Telugu Byte-Level BPE

Input Text: నందమూరి తారక రామారావు 50 సంవత్సరాలపైగా తెలుగు సినిమా రంగంలో కథా నాయకునిగా రాణించాడు

Tokens: ['న', 'ందమూరి', ' ', 'తారక', ' ', 'రామారావు', ' ', '50', ' ', 'సం', 'వత్సరాల', 'పైగా', ' ', 'తెలుగు', ' ', 'సినిమా', ' ', 'రంగంలో', ' ', 'కథా', '...']

7.6.2 Telugu Unicode-Level BPE

Input Text: నందమూరి తారక రామారావు 50 సంవత్సరాలపైగా తెలుగు సినిమా రంగంలో కథా నాయకునిగా రాణించాడు

Tokens: ['నందమూరి', 'తారక రామారావు', '50', 'సంవత్సరాల', 'పైగా', 'తెలుగు సినిమా', 'రంగంలో', 'కథా', 'నాయకు', 'నిగా', '...']

7.6.3 Telugu WordPiece

Input Text: నందమూరి తారక రామారావు 50 సంవత్సరాలపైగా తెలుగు సినిమా రంగంలో కథా నాయకునిగా రాణించాడు

Tokens: ['నందమూరి', 'తారక', 'రామారావు', '50', 'సంవత్సరాల', 'పైగా', 'తెలుగు', 'సినిమా', 'రంగంలో', 'కథా', 'న', 'గా', '...']

7.6.4 Bhojpuri Byte-Level BPE

Input Text: बहुत बढ़िया काम है यह।

Tokens: ['बहु', 'त', ' ', 'बढ', 'ि', 'िया', ' ', 'क', 'ाम', ' ', '...']

7.6.5 Bhojpuri Unicode-Level BPE

Input Text: भोजपुरी भारत के बिहार में बोली जाती है।

Tokens: ['भोजपुरी', ' ', 'भारत', ' ', 'के', ' ', 'बिहार', ' ', 'में', ' ', '...']

7.6.6 Bhojpuri WordPiece

Input Text: भोजपुरी भारत के बिहार में बोली जाती है।

Tokens: [भोजपुरी, भारत, के, बिहार, में, बोली, जाती, है,]

8 Technical Implementation

8.1 Data Pipeline Architecture

- **Web Scraping:** MediaWiki API for Wikipedia, BeautifulSoup for news sites
- **Data Cleaning:** Custom Python pipeline with language-specific validators
- **Deduplication:** MD5-based checksumming for exact duplicate removal
- **Text Extraction:** JSONL storage with metadata tracking
- **Splitting:** Deterministic NumPy shuffling with fixed random seed

8.2 File Structure

Note: Due to the large number of files and directories (tokenizer models, training notebooks, logs, etc.), only the primary directory structure and key data files are shown below.

```
telugu/
  configs/                # Configuration files & parameters
  data_collect/           # Data collection scripts
    scraper.py
    data_cleaner.py
    pipeline.py
  data/                   # Preprocessed dataset (17.4 GB total)
    config.json           # Complete statistics & metadata
    scrape_state.json     # Scraping checkpoint
    train/                # 20.3M lines (14 GB, 80%)
    val/                  # 2.5M lines (1.7 GB, 10%)
    test/                 # 2.5M lines (1.7 GB, 10%)
  model/                  # Model architecture & checkpoints
  tokenizer/              # Trained tokenizers (6 variants)
    full_byte_level/
    full_unicode_level/
    full_wordPiece_level/
  train/                  # Training scripts
  eval/                   # Evaluation scripts

bhojpuri/
  configs/                # Configuration files & parameters
  data_collect/           # Data collection scripts
    scraper.py
    data_cleaner.py
    pipeline.py
  data/                   # Preprocessed dataset (518 MB total)
    config.json           # Complete statistics & metadata
    scrape_state.json     # Scraping checkpoint
    train/                # 627.9K lines (414 MB, 80%)
    val/                  # 78.1K lines (52 MB, 10%)
    test/                 # 78.4K lines (52 MB, 10%)
  model/                  # Model architecture & checkpoints
  tokenizer/              # Trained tokenizers (3 variants)
    full_byte_level/
```

```
    full_unicode_level/  
    full_wordPiece_level/  
train/          # Training scripts  
eval/           # Evaluation scripts
```

9 Quality Assurance

9.1 Validation Checklist

- ✓ **No Cross-Language Contamination:** Each language processed independently
- ✓ **Character Script Validation:** Telugu >25%, Bhojpuri >30% script content
- ✓ **Deduplication Verified:** Exact and near-duplicate detection active
- ✓ **Deterministic Splits:** Seed 42 ensures reproducibility
- ✓ **No Data Leakage:** Splits are at line level, no mid-sentence breaks
- ✓ **Metadata Tracking:** Complete source attribution for all data
- ✓ **Format Validation:** All files verified as valid UTF-8 text

9.2 Reproducibility

All data processing is fully reproducible:

- **Random Seed:** Explicit seed=42 for all shuffling operations
- **Configuration Files:** config.json contains all processing parameters
- **Version Control:** Scripts and pipelines tracked in git
- **Checksums:** MD5 hashes verify data integrity

10 Challenges & Solutions

10.1 Challenge 1: Bhojpuri Data Scarcity

- **Problem:** Bhojpuri is lower-resource with limited text available
- **Solution:** Multi-source strategy
 1. Primary: HuggingFace Corpus (386K docs)
 2. Augmentation 1: OCR extraction from Hindi texts
 3. Augmentation 2: Machine translation (Hindi→Bhojpuri)
 4. Augmentation 3: Wikipedia dumps + web scraping
- **Result:** 776.8K lines collected (sufficient for Phase 2-3)

10.2 Challenge 2: OCR Quality

- **Problem:** Archive.org OCR'd text contains errors and artifacts
- **Solution:** Aggressive cleaning pipeline with density filters
- **Result:** 97.2% pass rate even on noisy OCR data

10.3 Challenge 3: Scale Management

- **Problem:** Telugu manual corpus is 15GB (21.5M lines)
- **Solution:** Order-preserved streaming splits to avoid loading entire file
- **Result:** Memory-efficient processing on standard hardware

11 Dataset and Tokenizer Repository

Phase 1 Data The complete Phase 1 dataset is available on Kaggle:

<https://www.kaggle.com/datasets/kspsvlnsiddardha/lma-slm>

11.1 Phase 1 Tokenizers

All 6 trained tokenizers (Telugu + Bhojpuri variants) are available on Kaggle:

<https://www.kaggle.com/datasets/kspsvlnsiddardha/lma-tokenizers>

11.2 Dataset Contents

- **Telugu Training Data:** 20.3M lines (80% of 25.3M total)
- **Telugu Validation Data:** 2.5M lines (10% of 25.3M total)
- **Telugu Test Data:** 2.5M lines (10% of 25.3M total)
- **Bhojpuri Training Data:** 627.9K lines (80% of 776.8K total)
- **Bhojpuri Validation Data:** 78.1K lines (10% of 776.8K total)
- **Bhojpuri Test Data:** 78.4K lines (10% of 776.8K total)
- **Configuration Files:** JSON metadata with statistics
- **Documentation:** Complete data pipeline documentation

A Configuration Files

A.1 Telugu config.json Structure

```
{
  "language": "Telugu",
  "data_sources": [...],
  "scraping": {
    "raw_texts": 32549,
    "texts_cleaned": 29206,
    "pass_rate_percent": 89.7
  },
  "splits": {
    "train": {"lines": 20281561, "percentage": 0.80},
    "val": {"lines": 2534020, "percentage": 0.10},
    "test": {"lines": 2533618, "percentage": 0.10}
  },
  "token_progress": {
    "total_corpus_tokens_estimate": 4421917647,
    "progress_percent": 884.3835
  }
}
```

A.2 Bhojpuri config.json Structure

Similar structure with Bhojpuri-specific sources and metrics (see main config file).

B References

- **Kaggle Dataset:** <https://www.kaggle.com/datasets/kspsvlnsiddardha/lma-slm>
- **Project Repository:** /media/ubuntu/Personal/IIIT Hyderabad/Semester 3/LMA/Mini_Project/
- **README:** README.md
- **Language Documentation:** bhojpuri/BHOJPURI.md, telugu/TELUGU.md
- **Configuration Files:** telugu/data/config.json, bhojpuri/data/config.json